Midterm 2 (practice version - NOT GRADED), Math 170a, Fall 2018
Instructor: Elizaveta Rebrova

Printed name: ______________________________

Signed name: ______________________________

Student ID number: ______________________________

Instructions:

- Read problems very carefully. If you have any questions please ask.
- The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems. Exception: Problem 1 (see instruction in the beginning of the statement).
- You final answers does not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- You are allowed to use only the items necessary for writing. Notes, books, sheets, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

1. For this problem only: please provide short answers (or short explanations when appropriate).

   (a) (2 points) Suppose that X is a random variable with the following PDF: $f(x) = cx^2$ when $1 \leq x \leq 2$ and 0 otherwise. Find c.
   Solution: Since we must have $\int_1^2 cx^2 = 1$, c satisfies $c2^3/3 - c1^3/3 = 1$. Hence, $7c/3 = 1$ and $c = 3/7$.

   (b) (2 points) If random variables X and Y are independent express $\text{Var}(X - Y)$ in terms of $\text{Var}(X)$ and $\text{Var}(Y)$.
   Solution: $\text{Var}(X-Y) = \text{Var}(X+(-Y)) = \text{Var}(X)+\text{Var}(-Y) = \text{Var}(X)+\text{Var}(Y)$.

   (c) (2 points) Let X be a random variable with expectation 0 and variance 1. Compute $\mathbb{E}(X^2 + X)$.
   Solution: Since $\mathbb{E}(X^2) = \text{Var}(X) + (\mathbb{E}X)^2 = 1 + 0^2 = 1$ we have $\mathbb{E}(X^2 + X) = \mathbb{E}(X^2) + \mathbb{E}(X) = 1 + 0 = 1$.

   (d) (2 points) Two random variables X and Y both have range $0, 1, 2$. Their joint PMF is given by $p_{X,Y}(x, y) = \frac{1}{3}y(x - 1)e^x$.
   Calculate $\mathbb{E}X$.
   Solution: This problem statement contains typos.

   There are multiple problems with given $p_{X,Y}(x, y)$ that you can recognize, including that $p_{X,Y}(0, 1) = \mathbb{P}(X = 0 \text{ and } Y = 1) = 1/2 \cdot 1(0-1)e^0 < 0$, and $\sum_{x,y} p_{X,Y}(x, y) \neq 1$.

   An admissible $p_{X,Y}(x, y)$ would be, for example, $p_{X,Y}(x, y) = \frac{1}{3e^2+3}(x - y)(x - 1)e^x$ (check it!)
   Then, $\mathbb{E}X = \sum_{x=0,1,2} xp_X(x) = p_X(1) + 2p_X(2) = \sum_{y=0,1,2} p_{X,Y}(1, y)$
   $+2\sum_{y=0,1,2} p_{X,Y}(2, y) = 0 + 2\frac{1}{3e^2+3}(2e^2 + 1e^2 + 0) = 2e^2/(e^2 + 1)$.

   (e) (2 points) Prove that X and Y from previous question cannot be independent. With corrected $p_{X,Y}(x, y)$: $\mathbb{P}(X = 2) \neq 0$ and $\mathbb{P}(Y = 2) \neq 0$, but $p_{X,Y}(2, 2) = \mathbb{P}(X = 2 \text{ and } Y = 2) = 0$, so X and Y are not independent.

2. (10 points) Let X_1, X_2, Y_1, Y_2 be four independent identically distributed random variables. If D denotes the distance between the points (X_1, Y_1) and (X_2, Y_2) show that

$$\mathbb{E}(D^2) = 4\,\mathrm{Var}(X_1).$$

Solution: Since $D^2 = (X_1 - X_2)^2 + (Y_1 - Y_2)^2$ we have

$$\begin{aligned}\mathbb{E}(D^2) = \mathbb{E}((X_1 - X_2)^2 + (Y_1 - Y_2)^2) &= \mathbb{E}(X_1^2 - 2X_1X_2 + X_2^2 + Y_1^2 - 2Y_1Y_2 + Y_2^2)\\ &\mathbb{E}(X_1^2) + \mathbb{E}(X_2^2) + \mathbb{E}(Y_1^2) + \mathbb{E}(Y_2^2) - 2\mathbb{E}(X_1X_2) - 2\mathbb{E}(Y_1Y_2).\end{aligned}$$

Since X_1, X_2, Y_1, Y_2 have the same distribution we have

$$\mathbb{E}(X_1^2) = \mathbb{E}(X_2^2) = \mathbb{E}(Y_1^2) = \mathbb{E}(Y_2^2),$$

and since they are all independent we have

$$\mathbb{E}(X_1X_2) = \mathbb{E}(X_1)\mathbb{E}(X_2) = \mathbb{E}(X_1)^2,\ \mathbb{E}(Y_1Y_2) = \mathbb{E}(Y_1)\mathbb{E}(Y_2) = \mathbb{E}(X_1)^2.$$

Therefore

$$\mathbb{E}(D^2) = 4\mathbb{E}(X_1^2) - 4(\mathbb{E}X_1)^2 = 4\,\mathrm{Var}(X_1).$$

3. (10 points) There are n couples coming to your dinner. These $2n$ people will be seated around a circular table which has exactly $2n$ seats in a uniformly random order. A couple will be happy if the partners are sitting next to each other or if there is exactly one person sitting between them.

   What is the probability that the first couple is happy?

   Compute the expected number of happy couples.

   Solution:

   - The first partner in the couple can sit in any of the $2n$ positions. But then there are only 4 positions for the 2nd partner. So they can be seated in $8n$ ways that they are happy. In total they can be seated in $2n(2n-1)$ ways. So the probability they are happy is
   $$\frac{8n}{2n(2n-1)} = \frac{4}{2n-1}.$$
   - Let X_i be the indicator variable of the event that i-th couple is happy. Then $\mathbb{E}(X_i) = \mathbb{P}(X_i = 1) = 4/(2n-1)$. The total number of happy couples is
   $$X = X_1 + X_2 + \cdots + X_n$$
   and the expectation
   $$\mathbb{E}(X) = \mathbb{E}(X_1) + \mathbb{E}(X_2) + \cdots + \mathbb{E}(X_n) = \frac{4n}{2n-1}.$$
   You can observe that this works for $n \geq 3$. For $n = 1, 2$ all the couple will be happy, so the answers for the first part are 1 and for the second part 1 (for $n = 1$) and 2 (for $n = 2$).

4. (10 points) We are given a random number X of fair coins. Assume that X has Poisson distribution with parameter 2. We toss each of the coins independently, and let Y be the number of heads that come up. Show that Y has Poisson distribution with parameter 1.

Solution: Given that $X = x$ the possible values for Y are $0, 1, \ldots, x$, and for $0 \le y \le x$ we have

$$p_{Y|X}(y|x) = \mathbb{P}(Y = y|X = x) = \binom{x}{y}\frac{1}{2^x}.$$

For $y > x$ we have $p_{Y|X}(y|x) = \mathbb{P}(Y = y|X = x) = 0$. Then

$$p_Y(y) = \mathbb{P}(Y = y) = \sum_x p_{Y|X}(y|x)p_X(x) = \sum_{x=y}^{\infty}\binom{x}{y}\frac{1}{2^x}e^{-2}\frac{2^x}{x!} = \frac{e^{-2}}{y!}\sum_{x=y}^{\infty}\frac{1}{(x-y)!} = \frac{e^{-1}}{y!},$$

which proves the claim.

5. (10 points) A factory has a machine that prints circuit boards. Each circuit board has a probability of p of jamming the machine. Once the machine jams, it destroys the circuit board that jammed it and stops working. Each circuit board contains a circuit with 6 components. In order to be working, at least 5 of these components must be functional. Each component is functional with probability a, independently.

   - What is the expected number of circuit boards printed (working or not, not including the destroyed board)?
   - What is the expected number of working circuit boards? Simplify your answer (no summations should be in the final answer).

   Solution: Let P be the number of circuit boards printed. Then since one is destroyed, $P + 1$ is a geometric random variable with parameter p. Thus $\mathbb{E}P + 1 = 1/p$, so

$$\mathbb{E}P = 1/p.$$

   First lets figure out the probability that any given board works. We have 6 independent components, each working with probability a and we want 5 or 6 to work, so the probability is

$$u := \binom{6}{5} a^5(1-a) + \binom{6}{6} a^6 = 6a^5 - 5a^6.$$

   Now lets figure out the conditional expectation $\mathbb{E}(W|P = n)$, where W is the number of working boards. If X_i is the random variable that is 1 if the ith printed board works and 0 otherwise then $X_i \sim$ Bernoulli(u). If we condition against $P = n$ then $W \sim X_1 + \ldots + X_n$ and W is binomial with parameters n and u. Thus $\mathbb{E}(W|P = n) = nu$. Using the formula for total expectation we get

$$\mathbb{E}W = \sum_{n=0}^{\infty} p_P(n)\mathbb{E}(W|P = n) = \sum_{n=0}^{\infty} unp_P(n) = u\mathbb{E}P =$$

$$u(\frac{1}{p} - 1) = (6a^5 - 5a^6)(\frac{1}{p} - 1).$$